

PROBLEM SET 4

Due date: April 12, 2013 at the recitation session

1. There are two players, a seller and a buyer. At date 1, the seller chooses his irreversible investment level $I \geq 0$ at cost I . At date 2, the seller may sell one unit of a good to the buyer, and the cost of supplying the good is $c(I)$ with $c' < 0$, $\lim_{I \rightarrow 0} c'(I) = -\infty$ and $c'' > 0$. To simplify, assume $c(0)$ is less than the buyer's valuation for the good, v . There is no discounting.
 - (a) What is the socially optimal level of investment?
 - (b) Suppose that at date 2 the buyer observes the investment made by the seller I and makes him a take-it-or-leave-it offer to buy the good for p . What is this offer? What is the subgame-perfect Nash equilibrium of the game?
 - (c) Can you think of a contractual way of avoiding the inefficient outcome of part b)? (Assume that contracts cannot be written on the level of I .)
2. Consider the following two stage model discussed in class. In the first stage, firms simultaneously choose whether to enter the market, resulting in a one time sunk cost of K for each entrant. In the second stage, the firms that chose to enter compete Cournot.
 - (a) Suppose that there are I identical consumers with individual demand functions $x_i = \alpha - \beta p$. Each firm has a constant marginal cost of production c . Derive the Cournot price and quantity as a function of the number of firms on the market J and the number of consumers I .
 - (b) Ignoring the integer problem, find the number of firms on the market that will choose to enter in the first stage in an SPNE of this game.
 - (c) What happens to the equilibrium number of entrants, the equilibrium price and quantity as $I \rightarrow \infty$? Explain.
3. Consider the following matrix game Γ :

		PI 2	
		Y_2	Z_2
PI 1	Y_1	0, 0	-1, 2
	Z_1	2, -1	0, 0

Figure 1

- (a) Does either player have a strictly dominant strategy in Γ ? If so, what is it?
- (b) Find all Nash equilibria (mixed and pure) of Γ .

Now consider the supergame $\Gamma^\infty(\delta)$, which consists of repeatedly playing Γ forever and in which the players possess the common discount factor $\delta \in (0, 1)$. Suppose that the players wish to support a SPNE in which they play (Y_1, Z_2) in even periods ($t = 0, 2, 4, \dots$) and (Z_1, Y_2) in odd periods ($t = 1, 3, 5, \dots$).

- (c) What is the discounted date-zero payoff to the players on the path suggested above?
 - (d) Find the value $\underline{\delta}$ such that the path suggested above can be supported in SPNE iff $\delta \geq \underline{\delta}$.
4. Suppose that there are two identical firms serving the market with inverse demand of

$$p = 1 - Q$$

where Q denotes the market output and p denotes the market price. Each firm's cost function is $C_i(q_i) = 0$ for $i = 1, 2$.

- (a) Consider a quantity-setting supergame. Calculate the lowest discount parameter $\underline{\delta}_C$ such that collusion on the monopoly output, in which the two firms share the earnings evenly can be supported in a SPNE with a trigger strategy of Cournot-Nash reversion.
- (b) Consider a symmetric price setting supergame. Calculate the lowest discount parameter $\underline{\delta}_B$ such that collusion on the monopoly price, in which the two firms share the earnings evenly can be supported in a SPNE with a trigger strategy of Bertrand-Nash reversion.